

SANJYOT SATISH AMRITKAR

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EDUCATION

Stony Brook University, Stony Brook, NY Aug 2024 – May 2026
MS Data Science GPA: 3.7/4.0 *Relevant: Algorithms, AI/ML, NLP, LLMs, Distributed Systems, Big Data, Statistics*

Savitribai Phule Pune University, Pune, India Aug 2019 – May 2023
BTech Computer Science GPA: 3.5/4.0 *Relevant: Data Structures, DBMS, Deep Learning, OS, Distributed Systems*

SKILLS

Languages: Python, Java, Go, C/C++, TypeScript, JavaScript (React.js, Node.js), SQL, R
AI & ML: PyTorch, TensorFlow, Scikit-learn, HuggingFace, LangChain, LangGraph, RAG, LLMs, Prompt Engineering
Backend & Data: REST APIs, FastAPI, Spring Boot, Microservices, Kafka, Spark, Pandas, ETL, SQLAlchemy
Tools & Platforms: Docker, Kubernetes, Git, CI/CD, Jenkins, Terraform, Tableau, Power BI, Azure AI Foundry
Databases & Cloud: PostgreSQL, MySQL, MongoDB, Redis, AWS (S3, Lambda, EC2, DynamoDB), GCP, Azure

PROFESSIONAL EXPERIENCE

Stony Brook University: AI Software Engineer (Research Assistant) Jun 2025 – May 2026

- Designed and shipped a production **AI-assisted ED triage platform** using **Python, FastAPI, React, PostgreSQL, Docker, and Azure**; deployed via **CI/CD** with zero downtime, used by clinical staff daily.
- Fine-tuned **Transformer NLP models (BERT, SentenceTransformers)** and LLM prompting workflows for intent classification and clinical summarization; improved classification accuracy by **30%**, reducing manual triage effort by **40%**.
- Implemented **LangChain and FAISS retrieval pipelines** over clinical guidelines to generate context-aware summaries; secured **service-to-service auth** and engineered concurrent session handling for high-volume traffic.

Nextracker: AI Software Engineering Intern Jun 2025 – Aug 2025

- Integrated **GPT-4, LLaMA 3, and Mistral** into enterprise products via **Azure AI Foundry**; reduced inference latency by **~30%** through prompt optimization, model selection, and infrastructure improvements.
- Designed and shipped **RAG-powered search and workflow automation** in **Go and React** using LangChain; eliminated **~40%** of manual troubleshooting time across production workflows.
- Benchmarked cloud vs. local LLM inference on accuracy, cost, and latency dimensions; findings informed the team's **production deployment strategy** and were documented as a **reusable technical resource** adopted as their long-term roadmap.

Barclays: Software Developer Jul 2023 – Jul 2024

- Developed **Java (Spring Boot)** REST APIs integrated with **AWS (S3, Glue, DynamoDB)**; automated backend data workflows processing **millions of financial records daily** with zero production incidents.
- Delivered an internal **React/Spring Boot** observability platform for application logging and diagnostics; improved debugging accuracy by **90%** and reduced mean time to resolution across distributed systems.
- Hardened **CI/CD pipelines** with SonarQube, Prisma, and JUnit; enforced quality gates at every stage, cutting production defect rates across a cross-functional Agile team.

Barclays: Software Engineering Intern Jun 2022 – Aug 2022

- Shipped a full-stack **Java, Spring Boot, and React** vulnerability dashboard on AWS; automated **API-database integration** cutting remediation time by **45%**.
- Participated in **Agile sprints, code reviews, and daily standups** alongside senior engineers; identified and resolved a critical **API defect** in the vulnerability tracking pipeline before sprint demo with no escalation.

PROJECTS

MediVault: AI-Powered Medical Claims Automation Dec 2025

- Developed a HIPAA-aware pipeline using **Python, OCR (Tesseract), and LLM-based parsing** on **AWS (Lambda, S3)** to extract structured data from medical documents; reduced manual data entry by **~60%**.
- Automated **ICD-10/CPT code recommendation** using local LLM inference (**Ollama**) at **~85% accuracy**; owned data ingestion, validation, inference, and **CMS-1500 claims generation** end-to-end with full **CI/CD deployment** on AWS.

CustomerPulse: Predictive ML and Analytics Platform Nov 2025

- Constructed an end-to-end ML pipeline using **Python, XGBoost, SHAP, SQL, and FastAPI** to identify high-risk customer segments; improved retention by **18%** through predictive modeling and explainable AI.
- Deployed real-time monitoring workflows with **Redis, ETL pipelines, and REST APIs**; reduced intervention time by **50%** through automated alerting and data quality validation.

E-commerce Product Chatbot using LLM and RAG May 2025

- Architected a full-stack AI application with a **FastAPI** backend, **FAISS** vector search, **SentenceTransformer** embeddings, and React interface; delivered **~40%** higher response relevance over keyword search.
- Exposed versioned **REST APIs** and deployed a containerized **FastAPI** service on **Kubernetes/GCP** with **CI/CD**; architected for **horizontal scalability** with clean service boundaries from day one.

ACHIEVEMENTS / EXTRACURRICULARS

President, IGSA @ Stony Brook · Python Tutor, CSTEP Program · Head Organizer, TEDxCCOEW